FAD-Net: Frequency-Domain Amplitude-Phase Decoupling Network for Optical-Elevation Remote Sensing Segmentation

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***Abstract*—Multimodal remote sensing semantic segmentation leverages multisource information for fine-grained scene analysis. However, due to the inherent limitations of spatial-domain representations, achieving effective feature interaction and reconstruction remains challenging. In this article, we propose the frequency-domain amplitude-phase decoupling network (FAD-Net). To resolve the fusion bottleneck caused by heterogeneous feature misalignment in the spatial domain, we design the multimodal Mona (MM-Mona) module. It establishes an interaction mechanism based on decoupled attributes recombination, physically decoupling textural and structural components to achieve complementarity between heterogeneous modalities. To address mutual interference between deep and shallow features in spatial decoding strategies, we propose the amplitude-phase recombination pyramid (APRP) decoder. A frequency-domain recombination mechanism is designed to reconstruct shallow details without losing deep semantic priors, guided by phase-geometric constraints. This effectively resolves the contradiction between preserving semantic consistency and recovering spatial details. Experimental results on the ISPRS Vaihingen and Potsdam datasets confirm that FAD-Net achieves state-of-the-art performance. The source code will be made available at https://github.com/Apsaras2005/FAD-Net**

***Index Terms*—Feature fusion, frequency-domain analysis, multimodal remote sensing, semantic segmentation.**

I. Introduction

MULTIMODAL remote sensing segmentation leverages multisource information—such as optical imagery and elevation data—to parse scenes into segmentation maps. It is widely applied in urban monitoring [1], land cover classification [2], precision agriculture, and disaster assessment [3].

Current multimodal methods fall into three categories: convolutional neural network (CNN)-based methods, Transformers, hybrid architectures, and advanced interaction mechanisms.

CNN-based methods employ parallel CNN backbones to extract features from different modalities. For feature complementarity, researchers use adaptive weighting or attention mechanisms [4], [5], [6]. Others introduce gating mechanisms to balance modality contributions and suppress interference from irrelevant information [7]. To enhance efficiency and feature representation, some works focus on compact cross-channel reconstruction and lightweight designs [8]. However, CNNs are constrained by local receptive fields, which limit their ability to model long-range dependencies.

The second category includes Transformers and hybrid architectures. Common strategies leverage cross-modal cross-attention or multiscale architectures to find second-order feature correlations [2], [9]. Given the limitations of single architectures, hybrid architectures combining CNNs and Transformers have become popular [10], [11]. They achieve complementary advantages through serial cascading or parallel encoding structures [12], [13], [14]. Meanwhile, to lower Transformer computational costs, researchers explore efficient parallel fusion and linear-complexity sparse attention [15], [16]. Furthermore, to reduce annotation reliance, self-supervised paradigms have been explored to mine robust representations via spatial–spectral masked autoencoders [17] and nearest neighbor-based contrastive learning [18]. Foundation model-prompted strategies [19], [20] also show promise by leveraging pretrained knowledge for weakly supervised scenarios [21], [22].

The third category focuses on advanced interaction mechanisms to address feature interference and semantic gaps. These employ semantic guidance strategies or build hypergraph structures to simulate high-order modal relationships and intuitive reasoning [23], [24]. Others introduce modality-specific experts, prototype-based information compensation [25] or cross-rectification modules to handle complex scenarios [26], [27], [28]. Some methods leverage category prior knowledge or geometric topology modeling to decouple feature fusion, suppressing modality-specific noise [29], [30], [31]. The selective scan mechanism based on the state space model (SSM), such as MSFMamba [32] and MGF-GCN [33], improves multimodal interaction efficiency by leveraging long-range modeling with linear complexity [34], [35], [36].

Unlike the methods above that focus on spatial-domain representations, some works use frequency-domain analysis for visual tasks. These methods leverage the frequency-domain global receptive field and spectral filtering to optimize features and suppress noise [1]. However, these studies are largely applied to single-modal data.

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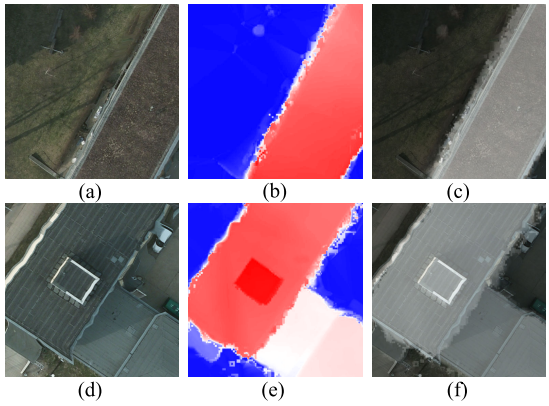


Fig. 1. Visualization of the fusion bottleneck caused by heterogeneous feature misalignment in the spatial domain. (a) and (d) Optical images. (b) and (e) Elevation data. (c) Disrupted geometric boundaries caused by optical shadow interference. (f) Suppressed texture details caused by the elevation surface smoothness.

While existing multimodal methods have advanced, due to the limitations of spatial-domain representations, they still face two challenges.

1) *Fusion Bottleneck Caused by Heterogeneous Feature Misalignment in the Spatial Domain:* Existing methods, including CNNs [4], [5], cross-attention [13], and gating strategies [7], [23], mostly use spatially weighted combinations. However, attribute entanglement is inherent to single-modal representations. Critical semantic features often couple with modality-specific redundant features. For example, optical texture details entangle with illumination-sensitive noise. Precise elevation geometry is entangled with sparse smooth surfaces. This representational discrepancy causes feature misalignment during spatial interaction. Specifically, as illustrated in Fig. 1, the structural integrity of elevation-derived boundaries is often disrupted by complex optical shadows acting as high-frequency noise, while rich optical textures are concurrently suppressed by the smooth surface characteristics of the elevation modality.
2) *Mutual Interference Between Deep Semantics and Shallow Details in Spatial Decoding:* Existing decoder designs, whether employing skip connections and cascaded upsampling [12] or pyramid modules [3], [35], largely employ spatial linear superposition to fuse deep and shallow features. Although these methods integrate multiscale information, their nonselective weighting is insufficient to handle representational discrepancies between deep and shallow features. Consequently, shallow features tend to disrupt the semantic consistency of deep features, while deep features smooth out the spatial details of shallow features.

To address the above challenges, we propose the frequency-domain amplitude-phase decoupling network (FAD-Net). Leveraging spectral complementarity, where the amplitude characterizes low-level statistical texture, and the phase encodes high-level geometric structures, we construct a frequency-domain interaction mechanism to decouple and fuse heterogeneous modalities. FAD-Net consists of two primary components. First, the encoder, with the multimodal Mona (MM-Mona) module, acts as a misalignment filter to resolve feature conflicts at the extraction stage. Second, the amplitude-phase recombination pyramid (APRP) decoder reconstructs spatial details guided by phase-geometric constraints while maintaining semantic consistency. Specifically, our main contributions are listed.

1) To address the fusion bottleneck caused by heterogeneous feature misalignment in the spatial domain, we propose the MM-Mona module, which constructs a feature-interaction mechanism based on decoupled attributes recombination. By decoupling and selectively exchanging textural and structural components in the frequency domain, it circumvents the spatial misalignment of heterogeneous features.
2) To address mutual interference between deep semantics and shallow details in spatial decoding, we introduce the APRP decoder. It creates a cross-level feature fusion mechanism based on frequency-domain attribute separation, circumventing the limitations of spatial linear superposition. Using the deep phase spectrum as a geometric prior to guide the reconstruction of the shallow features' amplitude spectra, it recovers spatial details while maintaining semantic consistency.

The rest of this article is organized as follows. Section II reviews the related work. Section III describes the architecture and key modules of FAD-Net. Section IV presents experimental results and analysis. Finally, Section V concludes the article.

II. Related Work

A. Visual Backbones and Fine-Tuning Strategies

Semantic segmentation backbones have evolved from CNNs to vision transformers (ViTs) [37]. Early CNN architectures, such as ABCNet [38], enhanced local feature extraction by using multiscale convolutions and attention mechanisms. However, due to their local receptive fields, CNNs struggle to capture global context. ViTs addressed long-range dependencies via attention mechanisms. Hierarchical architectures, such as Swin Transformer [39], [40], and SegFormer [41], improved multiscale representation and became mainstream for remote sensing segmentation.

However, full fine-tuning is increasingly expensive as models scale up. Parameter-efficient fine-tuning (PEFT) strategies, such as LoRA [42] and AdaptFormer [43], adapt models to specific tasks by training a small number of parameters. In visual tasks, standard adapters lack the capacity to handle complex spatial variations. To overcome this, Yin et al. [44] proposed Mona, an adapter specifically designed for visual tasks that utilizes multiscale depth-wise separable convolutions. Unlike linear adapters, it is more suitable for high-resolution semantic segmentation. We extend Mona to support multimodal frequency-domain interaction for efficient feature fusion.

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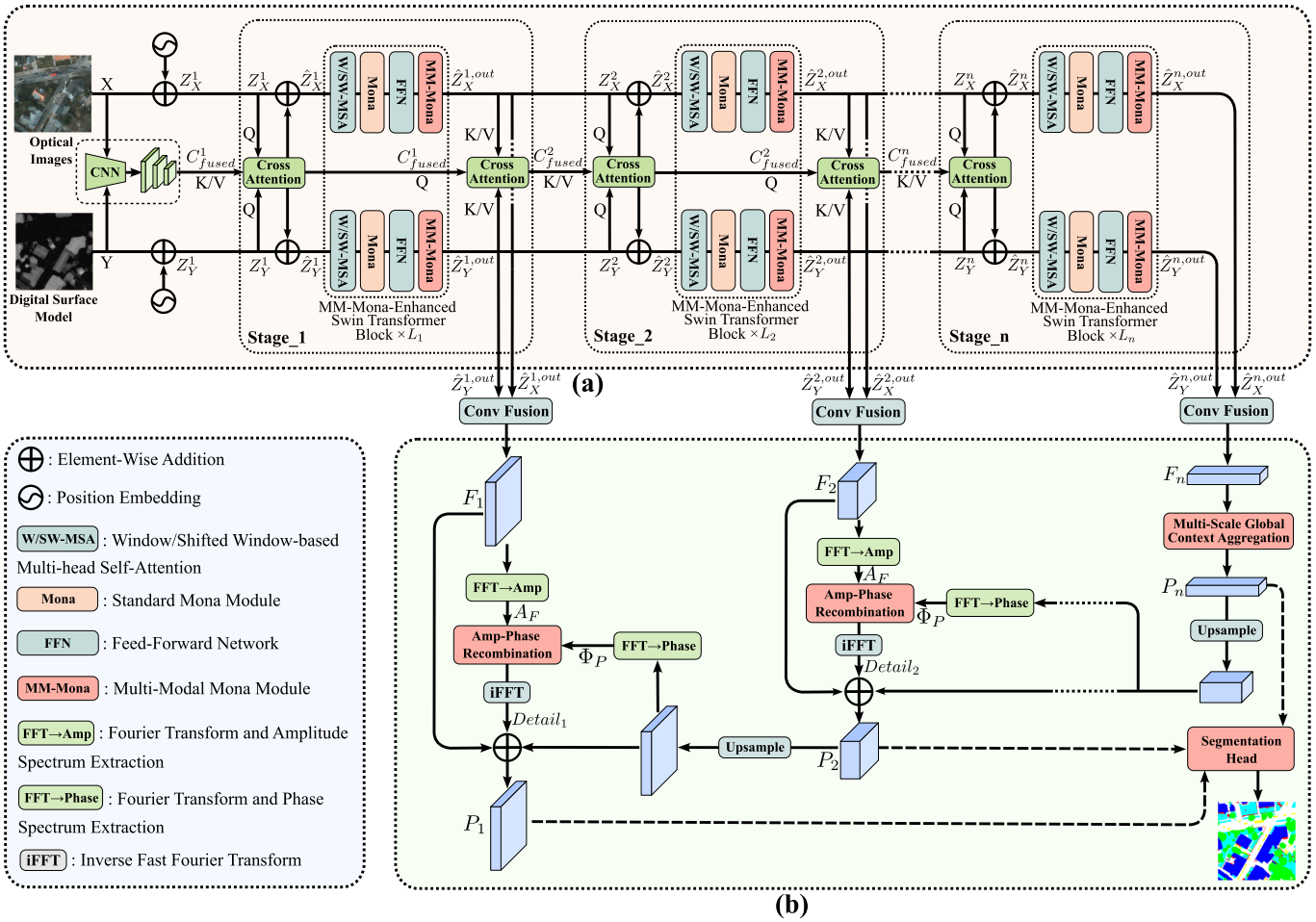


Fig. 2. Overall architecture of FAD-Net. The network utilizes a hierarchical encoder to extract multilevel features from optical images and the DSM. In each stage, the ViT-CNN bi-directional interaction structure captures local–global context, while the MM-Mona module performs frequency-domain decoupling and interaction to address feature misalignment. Consequently, the multiscale features are fed into the APRP decoder, which reconstructs shallow spatial details guided by deep phase-geometric priors to generate the final segmentation map. (a) Encoder. (b) Decoder.

B. Frequency-Domain Deep Learning Methods for Visual Tasks

Frequency-domain analysis, such as the fast Fourier transform (FFT), the discrete cosine transform (DCT), and the wavelet transform, has been introduced into deep learning [45]. AFENet [1] employs the FFT for adaptive frequency separation, generating masks that dynamically adjust high- and low-frequency components to supplement spatial features with fine-grained textures and global context. FDNet [46] combines Fourier and deformable convolutions, leveraging the frequency-domain global receptive field to extract road features. SFFNet [45] introduces the wavelet transform to decouple spatial features into high-frequency details and low-frequency approximations, achieving multiscale spatial–frequency fusion. FsaNet [47] focuses on efficiency by designing a frequency self-attention mechanism. By utilizing limited low-frequency coefficients to maintain high precision, it reduces computational complexity to a linear level.

Despite these advancements, significant gaps remain. Most existing works are limited to using frequency-domain analysis for single-modal feature enhancement [1], [45], [46]. Some studies also employ it as an auxiliary means to suppress noise in specific modalities [48] or reduce computational complexity [47]. However, exploring frequency-domain analysis for multimodal semantic segmentation to address heterogeneous data interaction and fusion challenges remains limited.

III. Proposed Method

Fig. 2 illustrates the proposed FAD-Net. FAD-Net is designed to process multimodal inputs, specifically optical images and digital surface model (DSM) elevation data. The encoder employs a ViT-CNN interaction structure and MM-Mona modules to extract and fuse multiscale features. These features are then fed into the APRP decoder, which reconstructs spatial details guided by deep geometric priors to generate the final segmentation map.

A. Overall Architecture

1) Encoder: The encoder comprises a ViT-CNN bi-directional interaction structure and MM-Mona-Enhanced Swin Transformer blocks.

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a) ViT-CNN bi-directional interaction structure: We incorporate the ViT-Adapter [49] to inject multiscale local spatial information into the ViT backbone, mitigating its lack of inductive biases. We first construct the initial multiscale feature vector C^1_{fused} by extracting multiscale CNN features $\{C^j_X\}^4_{j=2}$ and $\{C^j_Y\}^4_{j=2}$ from modalities X and Y, respectively,

$$C^1_{\text{fused}} = \text{Concat}^4_{j=2}\left(\text{Flatten}\left(\phi_{\text{fuse}}\left(C^j_X, C^j_Y\right)\right)\right) \tag{1}$$

where ϕ_{fuse} denotes a CNN fusion module, and Concat$(\cdot)$ represents the concatenation operation. At each stage i of the encoder, the ViT features Z^i_X and Z^i_Y of modalities X and Y interact bi-directionally with the CNN spatial prior C^i_{fused}. First, spatial information is injected into the ViT features

$$\hat{Z}^i_X = Z^i_X + \mathcal{A}\left(Q = P^i_{\text{proj}}\left(Z^i_X\right), K/V = C^i_{\text{fused}}\right) \tag{2}$$

$$\hat{Z}^i_Y = Z^i_Y + \mathcal{A}\left(Q = P^i_{\text{proj}}\left(Z^i_Y\right), K/V = C^i_{\text{fused}}\right) \tag{3}$$

where P^i_{proj} is a linear projection layer specific to stage i, and $\mathcal{A}$ denotes multiscale deformable attention. Subsequently, $\hat{Z}^i_X$ and $\hat{Z}^i_Y$ are processed by the Swin Transformer block $\mathcal{S}_i$ of the current stage to produce the outputs

$$\left(\hat{Z}^{i,\text{out}}_X, \hat{Z}^{i,\text{out}}_Y\right) = \mathcal{S}_i\left(\hat{Z}^i_X, \hat{Z}^i_Y\right). \tag{4}$$

Finally, we extract the ViT features to update the CNN spatial prior in a reverse direction

$$C^{i+1}_{\text{fused}} = \mathcal{A}\left(Q = C^i_{\text{fused}}, K/V_X = P^i_{\text{proj}}\left(\hat{Z}^{i,\text{out}}_X\right), K/V_Y = P^i_{\text{proj}}\left(\hat{Z}^{i,\text{out}}_Y\right)\right). \tag{5}$$

The updated C^{i+1}_{fused} serves as the CNN spatial prior for the subsequent stage.

b) MM-Mona-Enhanced Swin Transformer Block: As shown in Fig. 3, within each backbone Transformer block, we insert a standard Mona module [44] following the multi-head self-attention (MSA) layer to achieve parameter-efficient adaptation. We insert the MM-Mona module, detailed in Section III-B, after the feed-forward network (FFN) to perform deep cross-modal fusion and interaction in the frequency domain. Placing MM-Mona after the FFN ensures that fusion and interaction occur on high-level semantic representations rather than on raw, spatially mixed features post-MSA, allowing each modality to complete its independent encoding before cross-modal interaction.

2) Decoder: The outputs $\hat{Z}^{i,\text{out}}_X$ and $\hat{Z}^{i,\text{out}}_Y$ from various encoder stages collectively form a feature pyramid F

$$F = \left\{\phi_{\text{fuse}}\left(\hat{Z}^{i,\text{out}}_X, \hat{Z}^{i,\text{out}}_Y\right)\right\}^n_{i=1} \tag{6}$$

where ϕ_{fuse} represents a CNN fusion module. This pyramid is then fed into the proposed APRP decoder, detailed in Section III-C. To alleviate the class imbalance problem prevalent in remote sensing imagery, we utilize a hybrid loss function L_{total} to guide network optimization

$$L_{total} = \lambda_1 \cdot L_{CE} + \lambda_2 \cdot L_{Dice} \tag{7}$$

where L_{CE} denotes the cross-entropy loss, and L_{Dice} represents the dice loss. λ_1 and λ_2 are weighting coefficients.

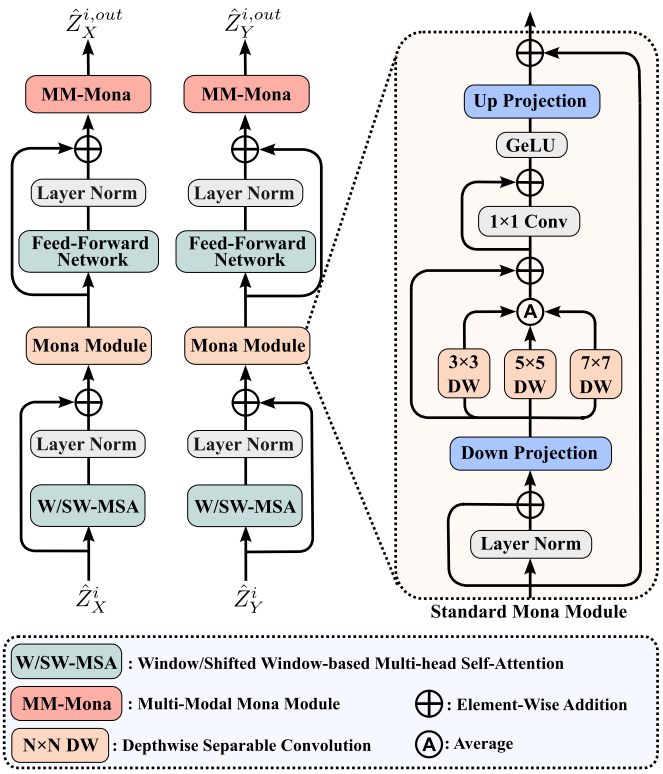


Fig. 3. MM-Mona-enhanced Swin Transformer block. A standard Mona module is embedded after the MSA for PEFT, while the MM-Mona module is embedded after the FFN to perform deep cross-modal fusion and interaction.

B. MM-Mona Module

Although the standard Mona adapter [44] leverages convolutional inductive biases, it processes modalities independently and aggregates multiscale features via static weights. Consequently, it fails to address spatial misalignment between heterogeneous features and scale variations. To address this, we propose the MM-Mona module, illustrated in Fig. 4. Unlike existing adapters, we introduce a parallel frequency branch to decouple modality-specific attributes via amplitude-phase interaction. Furthermore, to overcome static aggregation, we design the frequency-guided dynamic weight module (FGDWM) to adaptively adjust the spatial receptive field guided by spectral energy. Taking modality X as an example, we reshape the input feature $Z_X \in \mathbb{R}^{B\times N\times C}$ into $M_X \in \mathbb{R}^{B\times C_{\text{proj}}\times H\times W}$ via layer normalization and linear projection. Subsequently, it is fed into parallel frequency and spatial branches, bridged by the FGDWM.

1) Spatial Branch: The spatial branch utilizes dynamic weighting to address the content-agnostic limitation of the standard Mona adapter [44], which employs static averaging. Multiscale spatial feature aggregation is defined as

$$F^{\text{spatial}}_X = M_X + \sum_{n=1}^{N} w_n \cdot \mathcal{D}_n\left(M_X\right) \tag{8}$$

where M_X is the input feature; $\mathcal{D}_n$ represents the nth depth-wise separable convolution with a specific receptive field; w_n is the adaptive scalar weight generated by the FGDWM, detailed in Section III-B3; and N is the total number of branches.

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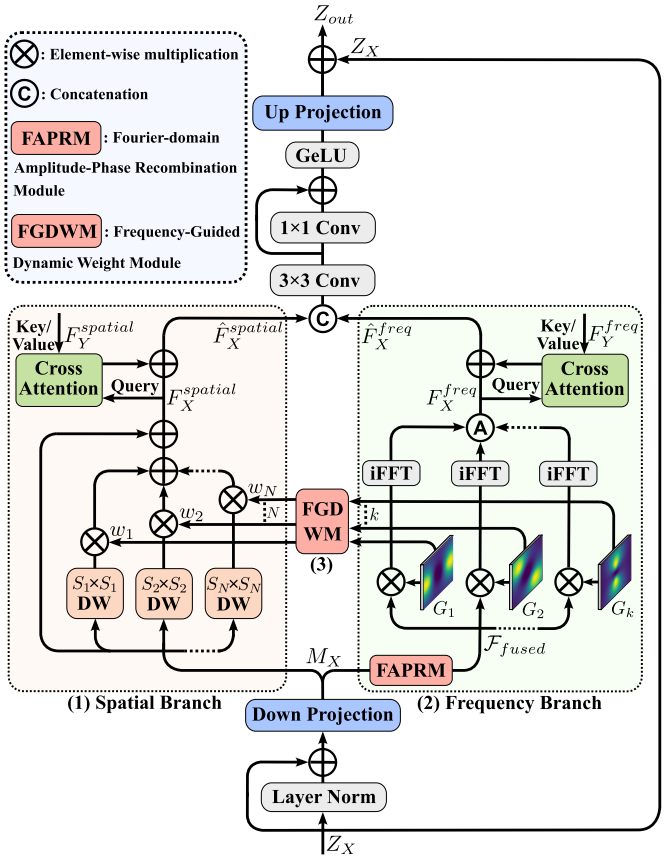


Fig. 4. MM-Mona module. This module consists of three components: 1) the spatial branch, responsible for the dynamic aggregation of multiscale features; 2) the frequency branch, which utilizes FAPRM to perform spectral decoupling and fusion; and 3) the FGDWM, which analyzes the frequency-domain energy distribution and generates adaptive weights to guide the selection of convolution kernels in the spatial branch.

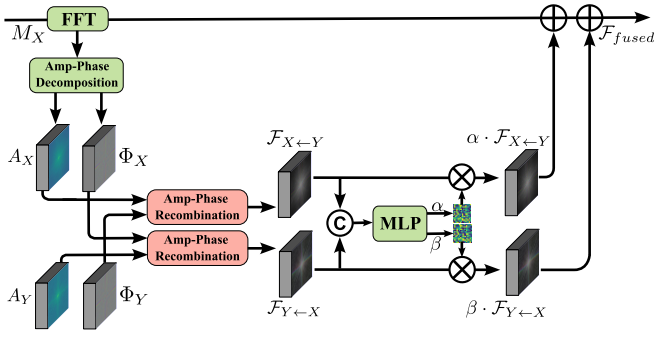


Fig. 5. FAPRM. This module decouples features into amplitude and phase components and adaptively fuses texture and structural information from different modalities by constructing a cross-modal recombined spectrum.

The generated spatial feature F_X^{spatial} subsequently interacts with the spatial feature of the other modality, F_Y^{spatial}, via cross-attention to obtain the enhanced spatial feature $\hat{F}_X^{\text{spatial}}$.

2) Frequency Branch: The frequency branch employs the Fourier-domain amplitude-phase recombination module (FAPRM) to execute a decoupling-recombination fusion strategy based on decoupled attributes, as shown in Fig. 5. To obtain frequency-domain decoupled feature representations, given spatial features M_X and M_Y, we first acquire their respective spectral components

$$A_X = |\mathcal{F}_c(M_X)| \tag{9}$$

$$\Phi_X = \angle\mathcal{F}_c(M_X) \tag{10}$$

$$A_Y = |\mathcal{F}_c(M_Y)| \tag{11}$$

$$\Phi_Y = \angle\mathcal{F}_c(M_Y) \tag{12}$$

where $\mathcal{F}_c$ denotes the shifted 2-D fast Fourier transform (2-D-FFT), A is the amplitude spectrum, and Φ is the phase spectrum. To mitigate the texture-structure ambiguity caused by direct spatial fusion of heterogeneous modalities, we construct a fused spectrum $\mathcal{F}_{\text{fused}}$ based on amplitude-phase exchange

$$\mathcal{F}_{\text{fused}} = \mathcal{F}_X + \alpha \cdot A_X \cdot e^{j\Phi_Y} + \beta \cdot A_Y \cdot e^{j\Phi_X} \tag{13}$$

where $A_X \cdot e^{j\Phi_Y}$ and $A_Y \cdot e^{j\Phi_X}$ are the reconstructed spectra by combining the current modality's amplitude spectrum with the other's phase spectrum. This leverages a Fourier property: the amplitude spectrum characterizes low-level statistics such as texture and illumination intensity. Conversely, the phase spectrum characterizes high-level geometric information such as edges and shapes. By performing cross-modal amplitude-phase recombination, we inject the textural features of one modality into the geometric topology of the other. This decoupled fusion strategy effectively circumvents feature interference and the disruption of geometric consistency caused by direct spatial fusion of heterogeneous modalities. The terms $\alpha, \beta \in \mathbb{R}^{B\times 1\times H\times W}$ are frequency-domain gating maps controlling the contribution of the cross-modal reconstructed spectra. They are constructed based on the cross-modal spectral discrepancies to capture nonoverlapping complementary regions rather than simple similarities

$$[\alpha, \beta] = \sigma\left(\phi_{\text{mlp}}\left(\text{Concat}\left[|A_X - A_Y|, |\Phi_X - \Phi_Y|\right]\right)\right) \tag{14}$$

where σ is the sigmoid function, and ϕ_{mlp} is a lightweight multilayer perceptron (MLP). Gates α and β adaptively enhance complementary spectral components while suppressing potentially conflicting ones.

Subsequently, $\mathcal{F}_{\text{fused}}$ is filtered by a bank of learnable Log-Gabor filters $G = \{G_k\}_{k=1}^{K}$. Log-Gabor filters are selected for their effectiveness in biological visual cortex simulation and texture analysis. The final frequency feature F_X^{freq} is calculated as

$$F_X^{\text{freq}} = \text{Re}\left(\frac{1}{K}\sum_{k=1}^{K}\mathcal{F}_c^{-1}\left(\mathcal{F}_{\text{fused}} \odot G_k\right)\right) \tag{15}$$

where $\mathcal{F}_c^{-1}$ denotes the inverse centralized Fourier transform, and $\odot$ represents element-wise multiplication. Since $\mathcal{F}_{\text{fused}}$ does not satisfy conjugate symmetry, the inverse transform yields complex-valued results. Consequently, the real part operator $\text{Re}(\cdot)$ maps features back to the real domain, ensuring compatibility with downstream network layers. Finally, F_X^{freq} interacts with the frequency feature of the other modality, F_Y^{freq}, via cross-attention to obtain the enhanced frequency feature $\hat{F}_X^{\text{freq}}$.

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3) Frequency-Guided Dynamic Weight Module: FGDWM establishes an adaptive mapping between frequency-domain energy distribution and spatial receptive field scales, generating the dynamic weights $[w_1, \ldots, w_N]$ required by the spatial branch. To obtain a frequency representation reflecting the global texture distribution, we first calculate the log-average amplitude spectrum S_{avg} of the input M_X

$$S_{\text{avg}} = \log\left(1 + \frac{1}{C_{\text{proj}}}\sum_{c=1}^{C_{\text{proj}}} |\mathcal{F}_c(M_X)_c|\right) \tag{16}$$

where $\mathcal{F}$ denotes the 2-D-FFT. To adaptively capture dominant frequency patterns in the current image, we introduce dynamic spectrum clustering. Treating S_{avg} as a frequency-domain energy density field, we identify a set of the most representative energy centroids $P \in \mathbb{R}^{N\times 2}$. To prevent centroid collapse into the low-frequency center caused by the $1/f$ spectral energy decay of natural images, P is initialized using a frequency-band prior strategy. Specifically, we evenly distribute the initial centroids across low-, mid-, and high-frequency bands via concentric anchor rings. The position update rule is defined as

$$P_j = \frac{\sum_{u\in\Omega_j} S_{\text{avg}}(u)\cdot u}{\sum_{u\in\Omega_j} S_{\text{avg}}(u) + \epsilon} \tag{17}$$

where P_j represents the coordinate of the jth centroid in set P, Ω_j denotes the Voronoi region associated with this centroid in the spectral domain based on the nearest neighbor principle, and u is the frequency coordinate vector within the region. This gradient-detached iterative process utilizes an energy-weighted mechanism to drive the centroids toward energy centers that accurately represent the dominant frequency patterns of the current feature.

To establish a physical connection between frequency-domain energy distribution and spatial-domain convolution kernel scale, we aggregate the Gabor texture response energy $E \in \mathbb{R}^K$ of the frequency branch into N energy centroids based on their center frequency coordinates $F \in \mathbb{R}^{K\times 2}$. Specifically, we calculate the membership degree of F to P and aggregate accordingly to generate grouped energy $E'_{\text{group}} \in \mathbb{R}^N$

$$M_{k,j} = \frac{\exp\left(-\|F_k - P_j\|_2\right)}{\sum_{n=1}^{N}\exp\left(-\|F_k - P_n\|_2\right)} \tag{18}$$

$$E'_{\text{group},j} = \sum_{k=1}^{K} E_k \cdot M_{k,j} \tag{19}$$

where $M_{k,j}$ indicates the probability that the kth Gabor energy belongs to the jth energy centroid. The final dynamic weights are generated as follows:

$$[w_1, \ldots, w_N] = \frac{\mathcal{S}\left(E'_{\text{group}}\right)}{\sum_{n=1}^{N}\mathcal{S}\left(E'_{\text{group}}\right)_n} \tag{20}$$

where $\mathcal{S}(E'_{\text{group}})$ denotes the result of sorting the grouped energy E'_{group} in ascending order based on the distance from the energy centroids to the origin. This establishes a spectral hierarchy from low-frequency structure to high-frequency texture. We align the spatial receptive field with the physical frequency scale by mapping low-frequency energy to large convolution kernels. This allows the model to capture long-range dependencies and global structures using large receptive fields. Conversely, high-frequency energy is mapped to small convolution kernels to focus on precise local detail extraction. This mechanism provides MM-Mona with content adaptivity, achieving the dynamic combination of multiscale spatial information based on intrinsic frequency characteristics.

After processing by the two parallel branches, the module output is

$$Z_{\text{out}} = Z_X + \mathcal{P}_{\text{out}}\left(\phi_{\text{fuse}}\left(\text{Concat}\left(F'^{\text{spatial}}_X, F'^{\text{freq}}_X\right)\right)\right) \tag{21}$$

where ϕ_{fuse} is a 3×3 fusion convolution, and $\mathcal{P}_{\text{out}}$ is the output projection that projects the features back to dimension C.

C. APRP Decoder

To address mutual interference between deep semantics and shallow details in spatial decoding, we propose the APRP decoder, as shown in Fig. 2.

The APRP decoder receives the feature pyramid $F = \{F_i\}_{i=1}^n$ from the encoder. The top-down fusion path originates from the deepest feature F_n. To capture global context, we first perform multiscale global context aggregation on F_n to generate the initial deep semantic feature P_n. For the ith level in the feature pyramid, we fuse the shallow feature F_i with the deep feature P_{i+1}. First, to acquire complementary information unique to features at different levels, we extract their specific spectral components

$$A_F = |\mathcal{F}_c(F_i)| \tag{22}$$

$$\Phi_P = \angle\mathcal{F}_c\left(\text{Up}\left(P_{i+1}\right)\right) \tag{23}$$

where $\mathcal{F}$ represents the 2-D-FFT, and $\text{Up}(\cdot)$ denotes bilinear interpolation upsampling. To construct a feature that possesses both rich texture and structural consistency, we utilize the decoupled spectral components to generate the detail feature $Detail_i$

$$\text{Detail}_i = \mathcal{P}\left(\text{Re}\left(\mathcal{F}_c^{-1}\left(A_F \cdot e^{j\Phi_P}\right)\right)\right) \tag{24}$$

where $A_F \cdot e^{j\Phi_P}$ represents the recombined spectrum generated by amplitude-phase recombination. The function $\mathcal{P}(\cdot)$ includes normalization and activation functions. Differing from spatial linear superposition, this strategy leverages geometric structure information from the deep feature phase spectrum to spatially guide the high-frequency texture amplitude components of shallow features. As a result, it preserves high-resolution details while suppressing potential background noise and semantic ambiguity. The output P_i is computed as

$$P_i = \phi_{\text{refine}}\left(Detail_i + \text{Up}\left(P_{i+1}\right) + F_i\right) \tag{25}$$

where ϕ_{refine} is a 3×3 convolution used for feature refinement and fusion. After iteratively generating pyramid features $\{P_i\}_{i=1}^n$ for all levels, the final segmentation map M is computed as

$$M = \phi_{\text{cls}}\left(\phi_{\text{fpn}}\left(\text{Concat}_{i=1}^{n}\left[\text{Up}\left(P_i \to P_1\right)\right]\right)\right) \tag{26}$$

where $\text{Up}(P_i \to P_1)$ indicates upsampling P_i to the size of P_1, Concat[·] is performed along the channel dimension, and

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ϕ_{fpn} and ϕ_{cls} denote the FPN bottleneck layer and classification head, respectively. Thus, the APRP decoder deeply integrates shallow high-frequency details and deep stable structures during top-down fusion, generating segmentation results with both correct semantics and sharp edges.

IV. Experimental Results

A. Datasets and Evaluation Metrics

We validated FAD-Net on ISPRS semantic segmentation benchmarks. Both datasets contain six categories: impervious surfaces, buildings, low vegetation, trees, cars, and clutter.

1) ISPRS Vaihingen: This dataset consists of 33 true orthophotography (TOP) tiles with an average 2494 × 2064 pixels. Each image contains three bands: near-infrared (NIR), red (R), and green (G), with a ground sampling distance (GSD) of approximately 9 cm. Following the standard benchmark partition, we used 17 tiles for testing and 16 for training.

2) ISPRS Potsdam: This dataset comprises 38 high-resolution images with a size of 6000 × 6000 pixels. We used the red (R), green (G), and blue (B) bands, and DSM data, with a GSD of approximately 5 cm. Standardly, we selected 14 tiles for testing and 24 for training.

We used three standard metrics: overall accuracy (OA), mean intersection over union (mIoU), and mean $F1$-score (mF1) to evaluate segmentation performance.

B. Implementation Details

1) Environment and Model Configuration: We implemented FAD-Net using PyTorch. Experiments were conducted on an NVIDIA GeForce RTX 4090 GPU. FAD-Net utilizes a hierarchical encoder with $n = 4$ stages. The number of stacked Swin Transformer Blocks in each stage was $[2, 2, 18, 2]$. We initialized the backbone using Swin Transformer V2 [50] weights pretrained on ImageNet-1K/22K. These weights remained frozen during training. Following the standard Mona [44], we configured $N = 3$ parallel convolution kernels with sizes of $S_1 = 3$, $S_2 = 5$, and $S_3 = 7$ in the spatial branch of the MM-Mona module.

2) Training Strategies: We employed data augmentation strategies, including random geometric transformations, multiscale random cropping, and color jittering. The input image size was set to 256 × 256. We set the loss function weighting coefficients $\lambda_1 = 0.8$ and $\lambda_2 = 0.2$. We employed the AdamW optimizer with a weight decay coefficient of 0.1. We set the initial learning rate to 1.5×10^{-4} and used a linear warmup for the first 10 epochs. Following warmup, the learning rate was decayed to 1.0×10^{-6} using a Cosine Annealing schedule. The training epochs were set to 100. All reported metrics are averages of five independent runs.

C. Hyperparameter Sensitivity Analysis

1) Impact of Log-Gabor Filter Number K: Table I shows significant performance improvement as K increases from 12 to 24. This suggests that sparse spectral sampling is insufficient to capture complex edge orientations and texture variations in remote sensing scenarios. However, mIoU declines slightly when K increases to 36. This is because excessive filters introduce feature redundancy and even amplify high-frequency noise and ringing artifacts.

TABLE I
Sensitivity Analysis of Log-Gabor Filter Number K

K	Vaihingen			Potsdam		
	OA	mIoU	mF1	OA	mIoU	mF1
12	0.9267	0.8456	0.9145	0.9198	0.8723	0.9307
24	**0.9290**	**0.8523**	**0.9188**	**0.9232**	**0.8785**	**0.9344**
36	0.9281	0.8490	0.9166	0.9220	0.8757	0.9327

TABLE II
Sensitivity Analysis of Loss Function Weights λ_1 and λ_2

Weights	Vaihingen			Potsdam		
	OA	mIoU	mF1	OA	mIoU	mF1
$\lambda_1 = 0.2, \lambda_2 = 0.8$	0.9271	0.8450	0.9142	0.9213	0.8767	0.9334
$\lambda_1 = 0.5, \lambda_2 = 0.5$	0.9259	0.8426	0.9127	0.9204	0.8756	0.9327
$\lambda_1 = 0.8, \lambda_2 = 0.2$	**0.9290**	**0.8523**	**0.9188**	**0.9232**	**0.8785**	**0.9344**

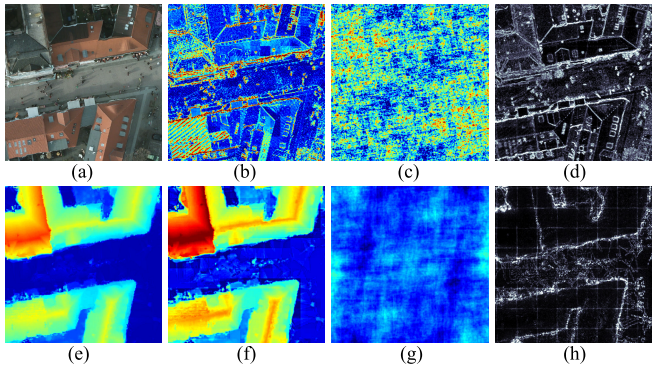


Fig. 6. Visual analysis of the frequency-domain amplitude-phase decoupling mechanism in FAPRM. (a) RGB image. (b) Feature map M_X of the RGB branch. (c) Feature map M_X reconstructed using only the amplitude spectrum. (d) Feature map M_X reconstructed using only the phase spectrum. (e) DSM image. (f) Feature map M_Y of the DSM branch. (g) Feature map M_Y reconstructed using only the amplitude spectrum. (h) Feature map M_Y reconstructed using only the phase spectrum.

2) Impact of Loss Weighting Coefficients λ_1 and λ_2: As shown in Table II, $\lambda_1 = 0.8$ and $\lambda_2 = 0.2$ achieve the best performance. Although Dice loss contributes to boundary optimization, giving it excessive weight leads to instability during the early stages of training due to significant gradient fluctuations, which severely disrupts the high-frequency structural consistency of the phase spectrum.

D. Mechanism Analysis

1) Effectiveness Analysis of FAPRM: As shown in Fig. 6, the amplitude spectrum preserves spectral intensity and texture patterns but lacks boundary information for ground objects. In contrast, the phase spectrum represents geometric structures

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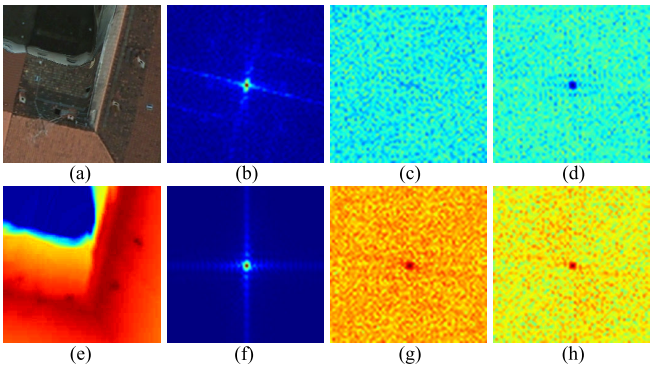


Fig. 7. Visual analysis of the frequency-domain amplitude-phase recombination mechanism in FAPRM. (a) RGB image. (b) Recombined spectrum $A_X \cdot e^{j\Phi_Y}$ generated by RGB amplitude and DSM phase. (c) Frequency-domain gating map α in the RGB branch for weighting the recombined spectrum $A_X \cdot e^{j\Phi_Y}$. (d) Frequency-domain gating map β in the RGB branch for weighting $A_Y \cdot e^{j\Phi_X}$. (e) DSM image. (f) Recombined spectrum $A_Y \cdot e^{j\Phi_X}$ generated by DSM amplitude and RGB phase. (g) Frequency-domain gating map α in the DSM branch for weighting $A_X \cdot e^{j\Phi_Y}$. (h) Frequency-domain gating map β in the DSM branch for weighting $A_Y \cdot e^{j\Phi_X}$.

TABLE III
DYNAMIC WEIGHT DISTRIBUTION GENERATED BY FGDWM IN DIFFERENT SCENES

Scene	RGB			DSM		
	w_3	w_5	w_7	w_3	w_5	w_7
1	0.4408	0.2050	0.3542	0.4262	0.2700	0.3038
2	0.1562	0.3116	0.5322	0.1651	0.3451	0.4898
3	0.1791	0.5318	0.2890	0.1637	0.3069	0.5294

that precisely outline the edge topology of ground objects. Such complementarity provides a physical basis for addressing feature misalignment of heterogeneous modalities in the spatial domain.

As shown in Fig. 7, the gating weight distribution reveals an asymmetric interaction pattern. The RGB branch gating map generally shows low responses and widespread suppression. In contrast, the DSM branch shows high-response activation. Specifically, the RGB branch tends to maintain lower gating weights to preserve the integrity of its original features. This prevents semantic confusion and spectral artifacts from excessive heterogeneous information. In contrast, the DSM modality is constrained by the limited information density of its feature space. Therefore, it exhibits a high dependence on complementary features. Meanwhile, the discrete and discontinuous gating map distributions on the spectral plane indicate that FAPRM performs a refined frequency selection.

2) Effectiveness Analysis of FGDWM: As shown in Fig. 8, the dynamic spectral clustering algorithm identifies low-frequency structural, medium-frequency texture, and high-frequency detail components. Table III presents the dynamic weight allocation results in three typical scenes. In Scene 1, the image contains small cars. Their edges manifest as abundant high-frequency components in the frequency domain. Consequently, both RGB and DSM branches allocate

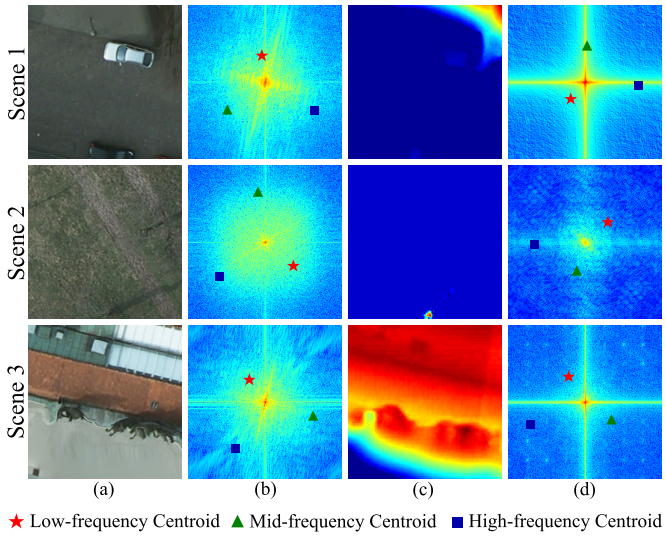


Fig. 8. Visualization of dynamic spectrum clustering results in FGDWM. (a) RGB image. (b) Log-amplitude spectrum of RGB features and its energy centroid clustering results; the marked points indicate the energy centers for low, medium, and high frequencies, respectively, where energy centroids further from the spectrum center represent higher frequency components. (c) DSM image. (d) Log-amplitude spectrum of DSM features and its energy centroid clustering results.

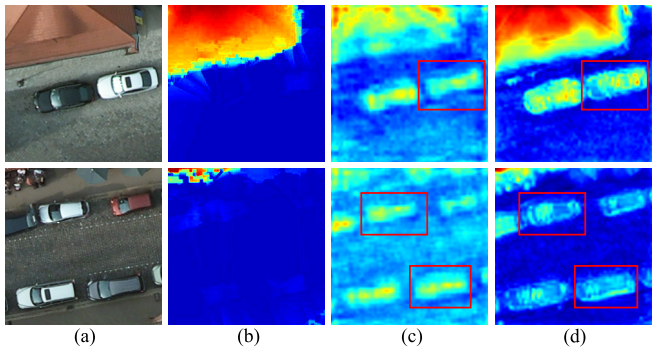


Fig. 9. Visual comparison between the feature fusion strategy of the APRP decoder and the traditional spatial-domain fusion strategy. (a) RGB image. (b) DSM image. (c) Fused feature map generated using a traditional spatial-domain linear superposition strategy. (d) Fused feature map generated using the proposed frequency-domain amplitude-phase recombination strategy.

the highest weights to small kernels. In contrast, Scene 2 comprises flat grassland with consistent texture, where spectral energy is primarily distributed in the low-frequency regime. Consequently, weight allocation favors large kernels.

Scene 3 shows a mix of buildings and roads. In the RGB branch, buildings and roads have regular textures, so medium-frequency energy is predominant. In contrast, in the DSM branch, since both buildings and roads appear as smooth surfaces, their representations are dominated by low-frequency components. This intermodal weight discrepancy demonstrates FGDWM can select the best spatial receptive field scale for each modality by leveraging inherent frequency properties.

3) Effectiveness Analysis of APRP Decoder: As shown in Fig. 9, traditional decoder architectures typically employ linear superposition in the spatial domain to fuse deep and shallow features, leading to mutual interference between deep

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TABLE IV
PERFORMANCE COMPARISON ON THE ISPRS VAIHINGEN DATASET

Method	IoU (%)					OA (%)	mF1 (%)	mIoU (%)
	Imp. surf	Buildings	Low veg.	Trees	Cars			
CMX [27]	85.98	93.16	69.69	80.47	80.48	91.23	89.88	81.96
PACSCNet [3]	86.58	93.91	70.83	82.96	76.79	91.96	90.03	82.21
MultiSenseSeg [26]	87.01	93.82	72.59	83.72	80.95	92.38	90.92	83.62
FTransUNet [12]	87.50	94.36	71.52	82.59	83.21	92.20	91.03	83.84
AFENet [1]	87.60	94.29	72.73	83.82	83.70	92.61	91.40	84.43
MFNet [51]	87.83	94.43	**73.68**	84.26	82.84	92.79	91.51	84.61
FAD-Net (Ours)	**88.14**	**94.73**	73.60	**84.30**	**85.39**	**92.90**	**91.88**	**85.23**

semantics and shallow noise. In contrast, the recombined feature maps generated by the APRP decoder effectively suppress background noise while preserving the clear contours of targets such as buildings and cars. This achieves a balance between high-frequency detail restoration and semantic structure preservation.

E. Comparative Experiments

We compared FAD-Net with six state-of-the-art multimodal methods.

1) Results on ISPRS Vaihingen Dataset: CMX [27] employs a dual-stream Transformer architecture to resolve information misalignment between RGB and DSM through a cross-modal feature rectification module. However, because it is inherently limited by weighted fusion in the spatial domain, it yields an mIoU of 81.96%, as shown in Table IV. To address detail loss during feature fusion, PACSCNet [3] introduces a progressive mechanism for adjacent-layer coordination. By enhancing interlayer feature interaction via symmetric cascade encoders, it improves edge capture capability and raises mIoU to 82.21%.

To balance computational efficiency and feature quality, MultiSenseSeg [26] introduces lightweight modality-specific expert modules that effectively mitigate interference from heterogeneous modalities, achieving an mIoU of 83.62%. FTransUNet [12] combines the local detail-extraction capability of CNNs with the long-range dependency-capture advantage of ViTs, achieving an mIoU of 83.84%. Although these methods have optimized fusion strategies, they predominantly rely on linear superposition in the spatial domain, failing to decouple modality-specific features from common features. With the introduction of foundational models, MFNet [51] uses the frozen SAM encoder and multimodal adapter fine-tuning. Benefiting from pretrained prior knowledge on massive natural scene data, this method shows a slight advantage in recognizing vegetation textures, whereas our phase-guidance inherently favors objects with regular geometric boundaries. The IoU for low vegetation is 0.08% higher than our method's, achieving an overall mIoU of 84.61%. The well-performing AFENet [1], while introducing frequency-domain masks to separate high- and low-frequency features, relies on simple binary truncation, which struggles to capture complex semantic meanings and is prone to introducing high-frequency noise.

Despite MFNet's superior performance on specific classes due to large-scale data priors, the proposed FAD-Net achieves the best results. By deeply integrating rich surface-texture information from infrared bands with precise geometric structure information from the DSM, it effectively resolves entanglement conflicts between heterogeneous modalities in the spatial domain. Particularly on the car category, which is highly susceptible to shadow occlusion, FAD-Net achieves an IoU of 85.39% by leveraging structural constraints from the phase spectrum, significantly outperforming PACSCNet (76.79%) and CMX (80.48%). Fig. 10 demonstrates that the segmentation maps generated by FAD-Net are superior to comparative methods in terms of object boundary integrity and semantic consistency.

2) Results on ISPRS Potsdam Dataset: Table V presents the results on the higher resolution ISPRS Potsdam dataset. CMX, constrained by its spatial rectification mechanism in high-resolution scenarios, has difficulty in distinguishing spectrally similar objects, resulting in an mIoU of 84.56%. PACSCNet leverages its cascade structure to preserve details layer by layer, raising the mIoU to 85.60%. MultiSenseSeg and FTransUNet reduce noise interference from small objects by enhancing context modeling, achieving mIoUs of 86.10% and 86.48%, respectively. These results indicate that in high-resolution scenes, contextual reasoning is indispensable beyond mere spatial detail recovery.

MFNet leverages the advantages of pretrained models for small-object recognition, achieving an mIoU of 86.93%. AFENet extracts edge details through adaptive frequency enhancement, achieving an mIoU of 87.43%. However, AFENet's mask-based frequency-separation strategy struggles to effectively differentiate features across different elevation planes. This is particularly true when confronted with similar vegetation textures in the Potsdam dataset. Consequently, its discriminability on tree and low vegetation categories is limited.

The proposed FAD-Net achieves the best performance with mIoU 87.85%. It is difficult to distinguish between low vegetation and trees because they share highly similar spectral features in RGB images. To address this, FAD-Net leverages geometric topology information in the DSM phase

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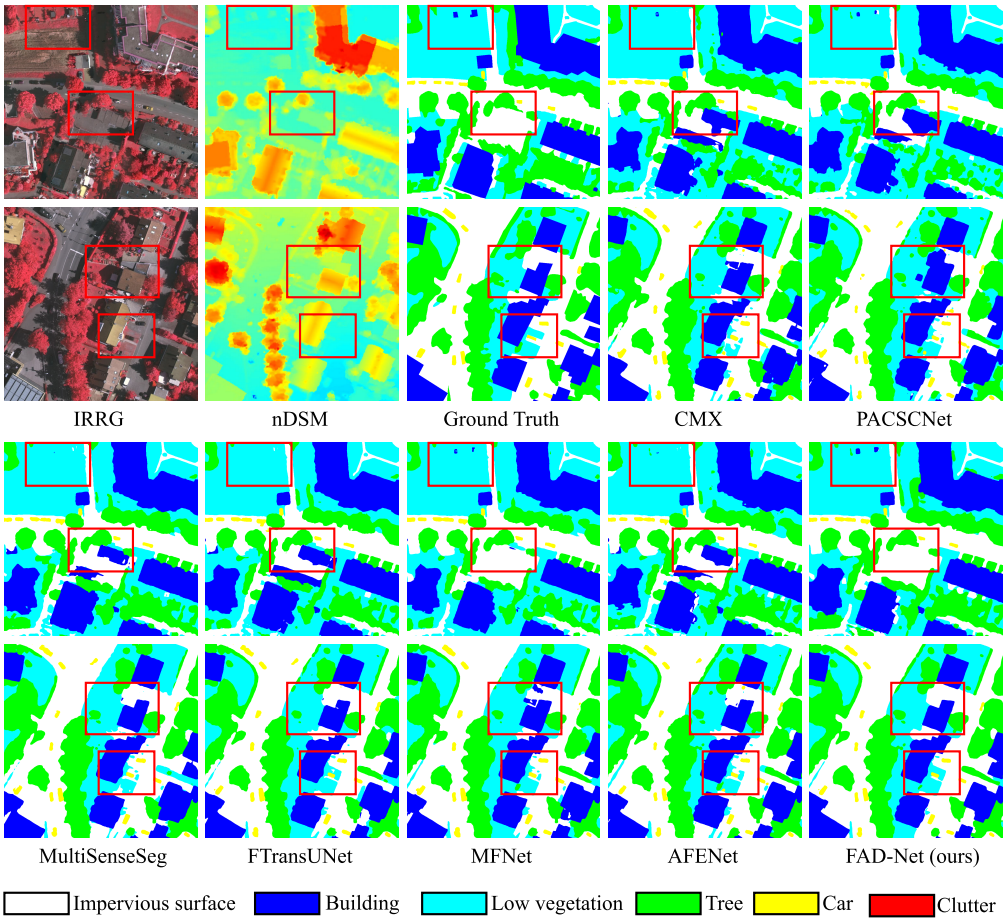


Fig. 10. Qualitative comparison of segmentation maps on the ISPRS Vaihingen dataset.

TABLE V
PERFORMANCE COMPARISON ON THE ISPRS POTSDAM DATASET

Method	IoU (%)					OA (%)	mF1 (%)	mIoU (%)
	Imp. surf	Buildings	Low veg.	Trees	Cars			
CMX [27]	85.79	93.64	75.97	76.79	90.59	90.29	91.47	84.56
PACSCNet [3]	86.15	93.70	77.95	79.04	91.18	90.88	92.12	85.60
MultiSenseSeg [26]	87.51	94.43	78.60	78.77	91.17	91.26	92.40	86.10
FTransUNet [12]	87.16	94.52	79.11	80.86	90.77	91.40	92.65	86.48
MFNet [51]	88.31	95.01	79.38	80.07	91.89	91.75	92.89	86.93
AFENet [1]	87.34	94.85	80.70	81.78	92.48	91.99	93.20	87.43
FAD-Net (Ours)	**88.46**	**95.05**	**80.99**	**82.18**	**92.58**	**92.32**	**93.44**	**87.85**

spectrum to guide the network toward correct discrimination via a cross-modal amplitude-phase recombination mechanism. Furthermore, thanks to MM-Mona's adaptive perception of high-frequency details, the model also realizes the best IoU on the car category. Fig. 11 demonstrates that FAD-Net is more robust when processing subtle textures and complex boundaries.

F. Ablation Studies

The baseline configuration consists of the standard Mona adapter [44] and the UPerNet decoder [52]. As shown in Table VI, by replacing UPerNet with the APRP decoder, we improved mIoU by 1.57% and 1.50% on the Vaihingen and Potsdam datasets, respectively. This demonstrates that using the deep phase spectrum as a structural guide to reconstruct the shallow amplitude spectrum outperforms spatial linear superposition.

Integrating the FGDWM into the baseline increased the mIoU by 0.89% and 1.16%. These results verify that establishing an adaptive mapping between frequency-domain energy distributions and spatial receptive fields significantly enhances multiscale spatial feature extraction. Meanwhile, adding the FAPRM alone resulted in improvements of 1.37 % and 1.51%.

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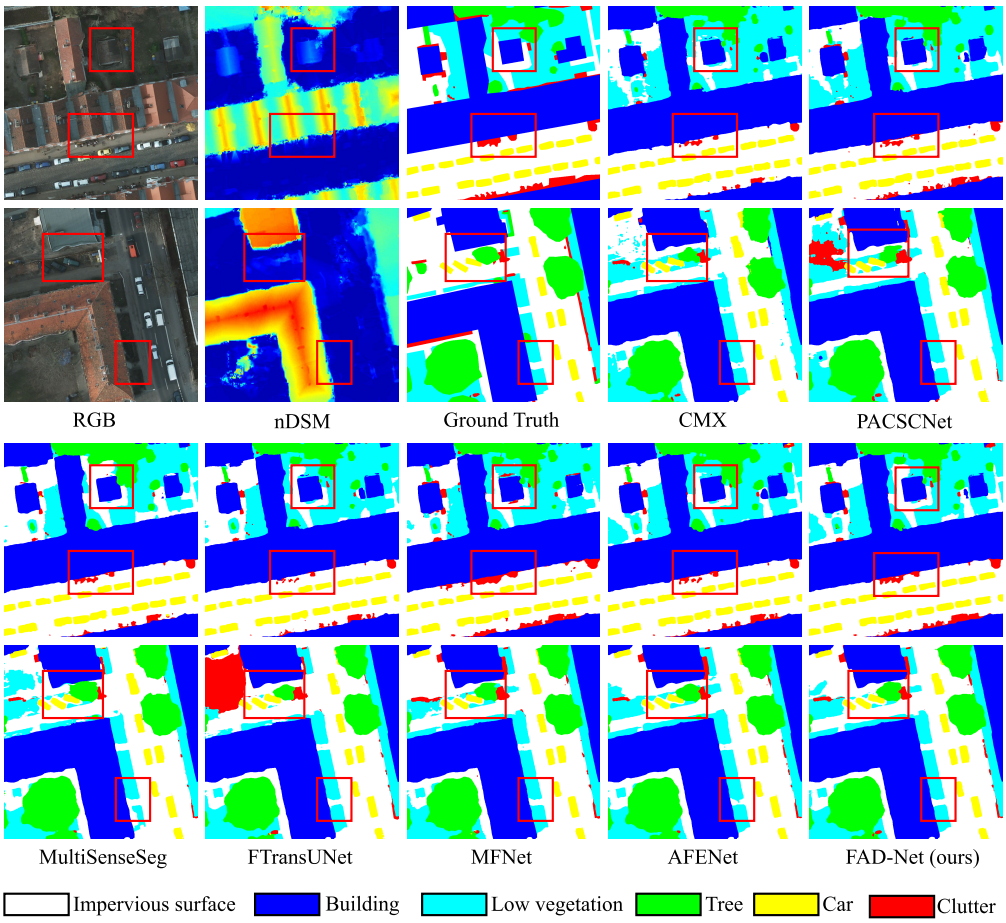


Fig. 11. Qualitative comparison of segmentation maps on the ISPRS Potsdam dataset.

TABLE VI
ABLATION STUDY OF KEY COMPONENTS ON ISPRS VAIHINGEN AND POTSDAM DATASETS

MM-Mona		APRP	Vaihingen			Potsdam		
FGDWM	FAPRM		OA	mIoU	mF1	OA	mIoU	mF1
			90.95	80.84	89.17	90.35	84.66	91.55
		✓	91.81	82.41	90.14	91.18	86.16	92.44
✓			91.42	81.73	89.72	90.94	85.82	92.24
	✓		91.84	82.21	90.02	91.28	86.17	92.45
✓	✓		92.20	83.30	90.69	91.67	86.71	92.76
✓	✓	✓	**92.90**	**85.23**	**91.88**	**92.32**	**87.85**	**93.44**

TABLE VII
MODEL COMPLEXITY AND EFFICIENCY ANALYSIS ON ISPRS POTSDAM DATASET WITH 256 × 256 INPUTS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	FLOPs (G)	Params (M)	Memory (MB)	FPS	mIoU (%)
CMX [27]	62.56	242.03	1002.52	20.52	84.56
PACSCNet [3]	47.97	128.97	590.40	22.44	85.60
MultiSenseSeg [26]	66.07	57.73	396.85	**29.57**	86.10
FTransUNet [12]	56.11	179.42	841.63	13.32	86.48
MFNet [51]	147.72	56.01	1306.14	8.15	86.93
AFENet [1]	**25.59**	**20.23**	**110.02**	16.18	87.43
FAD-Net (Ours)	40.93	133.09	1393.94	13.86	**87.85**

FAD-Net, integrating the MM-Mona module and APRP decoder, achieved mIoUs of 85.23% and 87.85%, outperforming the baseline. This indicates that the frequency-domain feature decoupling interaction at the encoder end and the feature recombination at the decoder end form a strong synergistic effect.

G. Model Complexity Analysis

As shown in Table VII, FAD-Net effectively balances computational overhead and segmentation accuracy. Compared to methods with more parameters, such as CMX [27] and FTransUNet [12], FAD-Net achieves higher segmentation performance with reduced parameters. To comprehensively evaluate the actual inference speed, we further report the frames per second (FPS). Due to the computational overhead introduced by FFT and complex-valued operations, FAD-Net achieves 13.86 FPS. While this is lower than that of lightweight models like MultiSenseSeg [26] (29.57 FPS), it is comparable to the complex hybrid architecture FTransUNet (13.32 FPS) and faster than the large-model-based MFNet [51] (8.15 FPS). This reflects a worthwhile tradeoff, where

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FAD-Net prioritizes superior segmentation precision for complex remote sensing scenarios while maintaining acceptable inference efficiency. While methods such as MFNet and MultiSenseSeg possess fewer parameters, their FLOPs are relatively high. In contrast, FAD-Net achieves lower FLOPs and competitive inference speed. However, parallel frequency branches and complex Fourier arithmetic increase GPU memory usage.

V. Conclusion

In this work, we proposed FAD-Net for the semantic segmentation of optical and elevation remote sensing imagery. Experiments verify that the MM-Mona module circumvents heterogeneous feature misalignment in the spatial domain. This reveals that decoupling heterogeneous features at the level of frequency-domain physical attributes is an effective pathway to break the multimodal fusion bottleneck. Meanwhile, the APRP decoder balances deep semantic consistency with shallow detail integrity. It provides a novel phase-geometry-guided frequency-domain recombination perspective to resolve mutual interference between deep and shallow features during decoding. However, due to the introduction of parallel frequency branches and complex-valued arithmetic, GPU memory consumption during training is relatively high. Future work will explore more compact and efficient frequency-domain computing strategies and attempt to extend this frequency-domain decoupling idea to more complex mul timodal Earth Observation scenarios.

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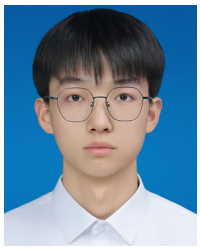
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